Backbones and No Backbones · Teacher and Cover Plan

Learning goal: I can find a backbone on a diagram and sort animals with and without a backbone.

Scheme Outcome

BASE Science scheme of work, Unit 1 (Skeletons, Muscles and Nutrition), Topic: Compare animals with and without a backbone (vertebrates and invertebrates).

Compare animals with and without a backbone and sort them using body clues.

Success Criteria

I can point to a backbone on an inside picture.

I can sort animals into backbone and no backbone.

I can name one way an animal with no backbone stays supported.

Model Sequence (I do)

Look at the fish diagram first. It shows a line of small bones down the middle.

When an animal has a backbone inside, we put it in the backbone group.

Now look at the crab. Its hard case is outside, and it has no backbone inside.

Do not guess from size. A tiny mouse has a backbone inside.

Do not guess from legs. A fish has no legs, yet it has a backbone.

An animal with no backbone can still have support, like a shell or a soft wet body.

We Do & Hinge Check

Which clue proves an animal has a backbone?

Answer: A. A line of small bones down the back proves it has a backbone.

Safe Practical & Card Setup

Sort the animal clue cards into two groups. Pick one animal and show its body clue.

Kit: Printed animal x-ray cards, two sorting mats (Backbone / No backbone), and word bank strips.

Use the printed cards on a flat desk. Keep walkways clear. Do not press on anyone's back.

You Do Routes (Supported, Independent, Aspire)

Supported: Sort four animal cards onto the two mats with help. Point to the bone row. (Help: Choose has a backbone or no backbone. Point to the inside clue.)

Independent: Sort six animal cards into two groups. Pick one animal and write a because sentence. (Help: The __ has / does not have a backbone because I can see __.)

Aspire: Compare a small mouse and a large crab. Explain why size does not tell us the group. (Help: A tiny mouse has a __ inside, while a big crab has a __

outside.)